Code No: 123BN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, September/October - 2023
MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a) List the applications of propositional logic. [2]
b) Write Converse, opposite for the statement "If 2+2=8 then sun rises in the east". [3]
c) Define a Function. [2]
d) Represent the following relation R on the set X={1,2,3} in matrix form and graph form R = {(1,1),(2,2),(3,3),(2,3),(3,1),(1,3)}. [3]
e) List the applications of Pigeonhole principle. [2]
f) How many 4-digit numbers are possible with the digits ***1,3,4,6,7,9*** if repetition of digits are allowed. [3]
g) A sequence is given as 1,1,1,2,2,2,3,4,4,5. Compute generating function for this sequence. [2]
h) Solve the following recurrence relation by substitution method : $\mathbf{a_n = a_{n-1} + n.}$ [3]
i) A connected graph G contains 5 vertices and 10 edges. What is the number of edges that has to be deleted from G to form a tree? [2]
j) Graph can be represented with adjacency matrix and adjacency list. Give differences between these two methods. [3]

PART - B

(50 Marks)

2.a) Convert $(P \rightarrow Q) \wedge (P \rightarrow \neg Q)$ in principal conjunctive normal form.
b) Show that $R \rightarrow S$ can be derived from the premises $P \rightarrow (Q \rightarrow S)$, $\neg R \vee P$ and Q. [5+5]

OR

3.a) Derive $P \rightarrow \neg S$ from $P \rightarrow (Q \vee R)$, $Q \rightarrow \neg P$, $S \rightarrow \neg R$, P by applying inference rules.
b) Obtain Principal Conjunctive Normal Form (PCNF) for the following formula: $(P \vee Q) \wedge (\neg Q \vee R) \wedge (Q \rightarrow R)$. [5+5]

4.a) Draw the Hasse diagram for the divisibility on the set {1,2,3,6,12,24,36,48,96}.
b) Consider the following relation R on the set {1, 2, 3, 4}. Determine whether it is reflexive, irreflexive, symmetric, anti-symmetric and transitive : R={(1,1),(2,2),(2,3),(2,4),(3,2),(3,3),(3,4)}. [5+5]

OR

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5.a) Let X = {1,2,3,4,5,6} and relation R on set X is defined as R = {(a,b) | a-b is divisible by 3} . Show that R is an equivalence relation.

b) Show that the relation R = {(a,b) | if a divides b} on set X={1,2,3,4,6,8,12,16,32,36} is partial ordered relation and draw Hasse diagram for it. [5+5]

6.a) Find the number of ways of distributing 50-similar balls into 4 distinct boxes such that first two boxes should contain atleast two balls and next two boxes must contain atleast three ball.

b) List 3-combinations and 3-permutations of {4a,5b,1c,2d}. [5+5]

OR

7.a) Expand the multinomial $(2x_1+4x_2+x_3)^4$.

b) Find the number of integral solutions to $x_1+x_2+x_3+x_4 = 50$, where $x_1 \geq 4$, $x_2 \geq 7$, $x_3 \geq 1$, $x_4 \geq 10$. [5+5]

8.a) Solve the following recurrence relation using generating function: $a_n - 5\,a_{n-1} + 6\,a_{n-2} = 0$ for $n \geq 2$, and $a_0 = 1$ and $a_1 = -2$

b) Solve the following recurrence relation $F_n = F_{n-1} + F_{n-2}$, where $F_0=1$, $F_1=1$. [5+5]

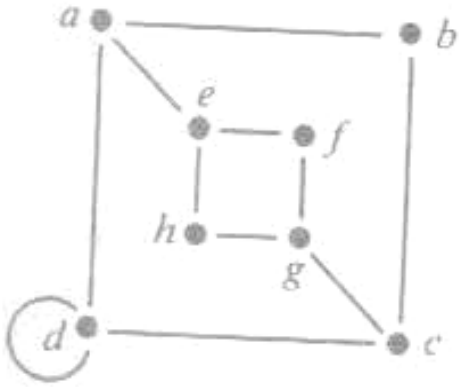
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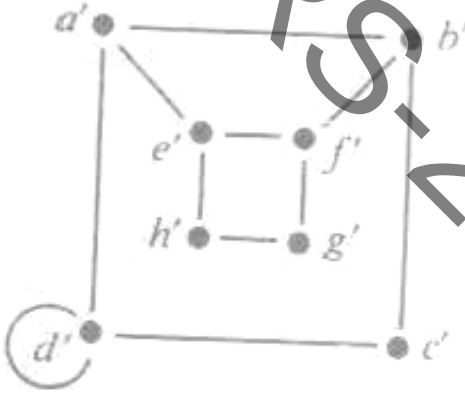
9.a) Solve $a_n^{\,2} - 2a_{n-1} = 1$ for $n \geq 1$, where $a_0 = 2$.

b) Find a general expression for a_n using generating functions. $a_n - 4a_{n-1} + 3a_{n-2} = 0, n \geq 2$, $n \geq 2$. [5+5]

10.a) Verify the graphs $\mathbf{K_5}$ (Complete Graph with 5 vertices), $\mathbf{K_4}$(Complete Graph with 4 vertices), $\mathbf{K_{4,3}}$ (Complete Bi-Partite Graph), $\mathbf{K_{2,44}}$(Complete Bi-Partite Graph) are planar graphs or not? Explain.

b) Verify whether the following two graphs are isomorphic or not? Justify your answer. [5+5]

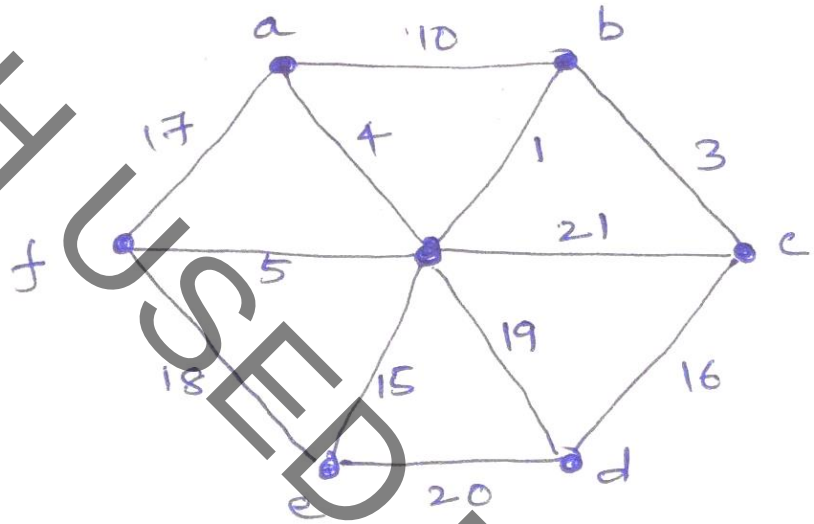




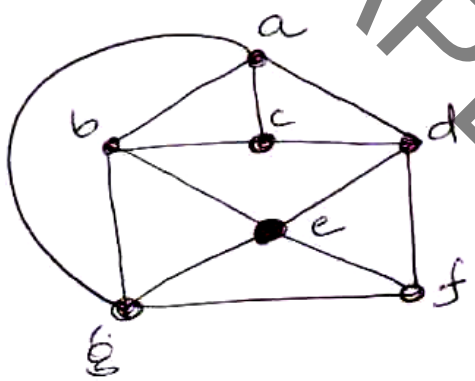
OR

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11.a) Apply Kruskal's algorithm and Prime's algorithm to find minimal spanning tree on the following graph and compare these two methods.



b) Define Chromatic number of a graph. Compute the chromatic number of the following graph. [5+5]



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